

PHYS 230 - Electricity and Magnetism

Lecture 7

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

1 Electric Charge and Electric Field

- Electric Dipoles

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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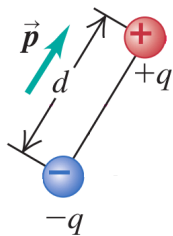
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Electric Dipoles

Electric dipole: a pair of point charges with

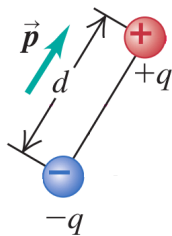
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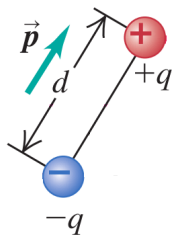
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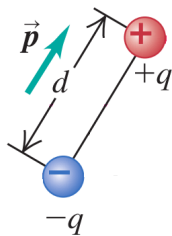
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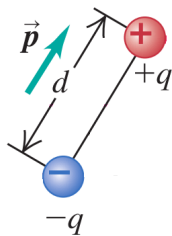
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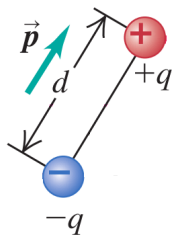
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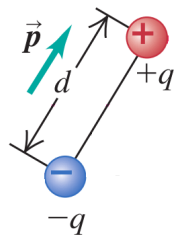
- **Magnitude of $\mathbf{p}$:**

$$p = qd$$

Electric Dipoles

The units of electric dipole moment is

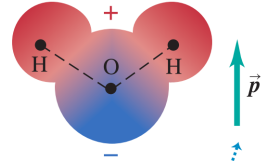
$$[\mathbf{p}] = [q] [\mathbf{r}_{-+}] = \text{C} \cdot \text{m}$$



Electric Dipoles: Water Molecule

Water molecule is overall neutral but has a polarization and is a dipole!

(a) A water molecule, showing positive charge as red and negative charge as blue



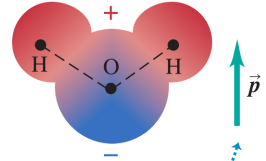
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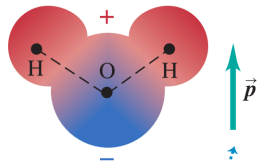
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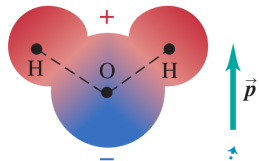
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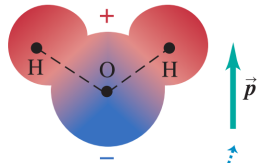
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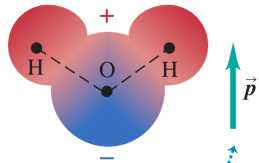
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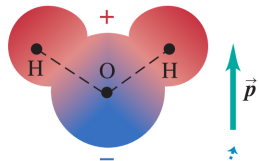
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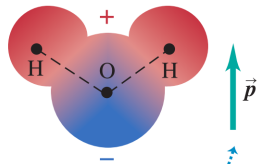
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- Our very existence as a living being depends on electric dipoles!

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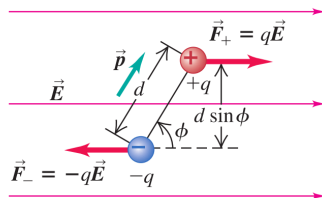
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Forces and Torques on Electric Dipoles

Let us place a dipole in a uniform electric field (as in picture)

- Force $\vec{F}_+$, $\vec{F}_-$ on q and $-q$: same magnitude, opposite directions

Figure 21.31 The net force on this electric dipole is zero, but there is a torque directed into the page that tends to rotate the dipole clockwise.

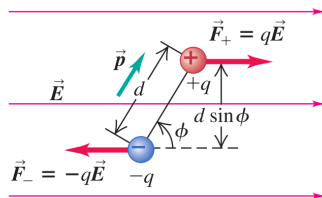


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 - The net force $\vec{F} = \vec{F}_+ + \vec{F}_-$ on an electric dipole in a uniform external electric field is **zero**

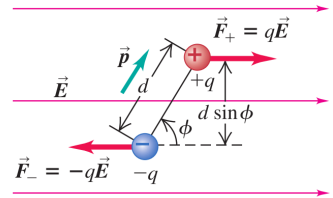
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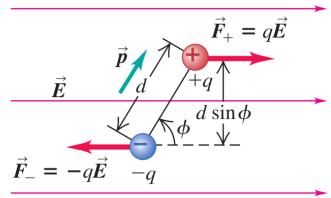
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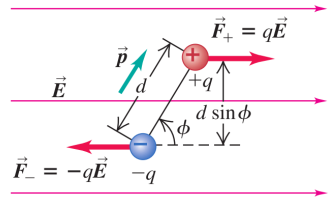
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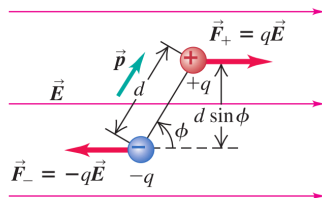
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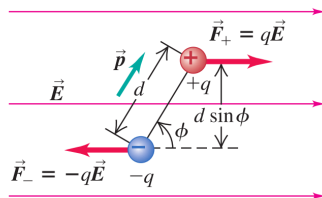
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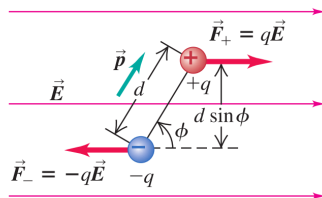
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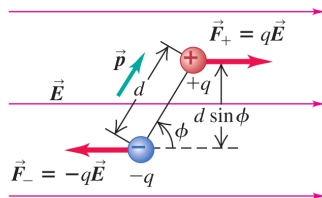
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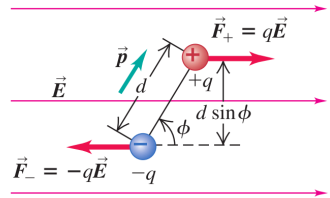


Forces and Torques on Electric Dipoles

- The magnitude of the torque on the dipole is

$$\begin{aligned} |\boldsymbol{\tau}| &= |\mathbf{p} \times \mathbf{E}| \\ &= pE \sin(\phi_{\mathbf{p}, \mathbf{E}}) \\ &= qdE \sin(\phi_{\mathbf{p}, \mathbf{E}}) \end{aligned}$$

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- The direction of τ is found using the right-hand rule

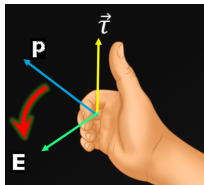
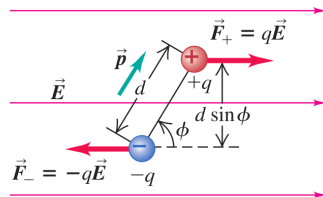


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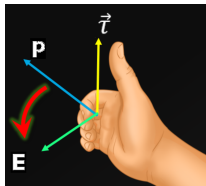


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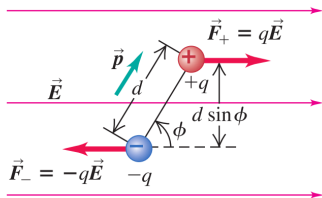
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- The units of τ is

$$[\tau] = [\mathbf{p}] [\mathbf{E}] = [q] [\mathbf{r}] [\mathbf{E}] = \text{N} \cdot \text{m}$$

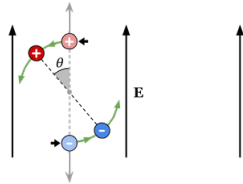
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Potential Energy of Electric Dipoles

The potential energy of a dipole with moment $\mathbf{p}$ placed in an external field $\mathbf{E}$, is the negative of the work done by $\mathbf{E}$ on the dipole

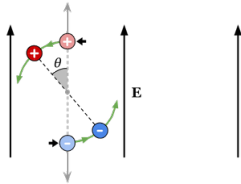
$$dU = -dW$$



Potential Energy of Electric Dipoles

Torque τ that rotates the dipole by an angle $d\theta$ does work as

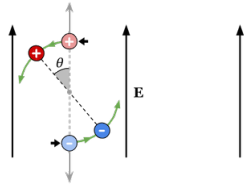
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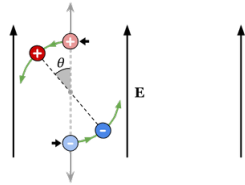
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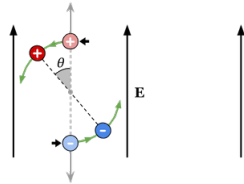
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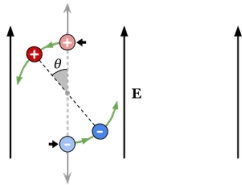
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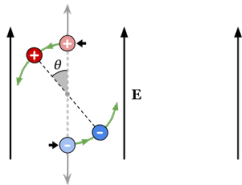
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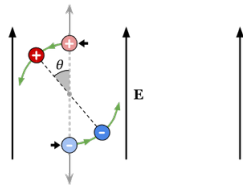
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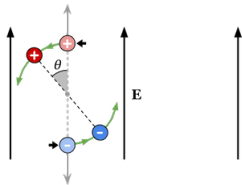
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Potential Energy of Electric Dipoles

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$$\begin{aligned}dW &= dW_+ + dW_- \\&= \mathbf{F}_+ \cdot d\mathbf{r}_+ + \mathbf{F}_- \cdot d\mathbf{r}_- \\&= q\mathbf{E} \cdot d\mathbf{r}_+ + (-q)\mathbf{E} \cdot d\mathbf{r}_- \\&= q\mathbf{E} \cdot (d\mathbf{r}_+ - d\mathbf{r}_-)\end{aligned}$$



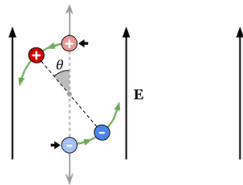
For infinitesimal $d\mathbf{r}$, we can assume $\mathbf{E}$ is constant to a good approximation, and hence the total work is

$$\begin{aligned}W &= \int dW = q\mathbf{E} \cdot \int (d\mathbf{r}_+ - d\mathbf{r}_-) \\&= q\mathbf{E} \cdot (\mathbf{r}_+ - \mathbf{r}_-) \\&= q\mathbf{E} \cdot \mathbf{r}_{-+} \\&= q\mathbf{r}_{-+} \cdot \mathbf{E}\end{aligned}$$

Potential Energy of Electric Dipoles

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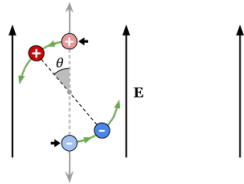
Potential Energy of Electric Dipoles

So since

$$\int dU = - \int dW$$

we obtain the potential energy of a dipole with moment $\mathbf{p}$ placed in an external field $\mathbf{E}$ as

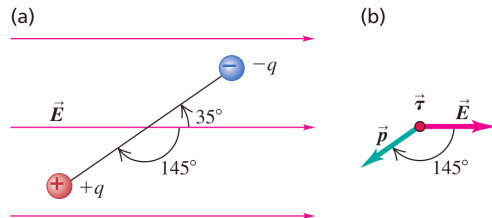
$$U = -\mathbf{p} \cdot \mathbf{E}$$



Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

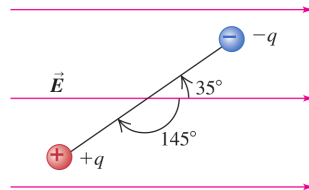
Figure shows an electric dipole in a uniform electric field of magnitude $5 \times 10^5 \text{ N/C}$ that is directed parallel to the plane of the figure. The charges are $\pm 1.6 \times 10^{-19} \text{ C}$, both lie in the plane and are separated by $1.25 \times 10^{-10} \text{ m}$. Find (a) the net force exerted by the field on the dipole; (b) the vector electric dipole and its magnitude; (c) the torque vector and its magnitude and direction; (d) the potential energy of the system in the position shown.



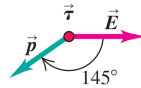
Exercise I

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(a)



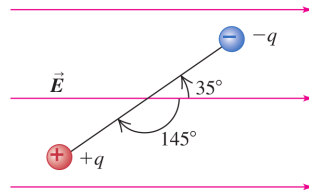
(b)



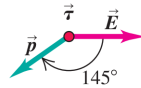
Exercise I

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(a)



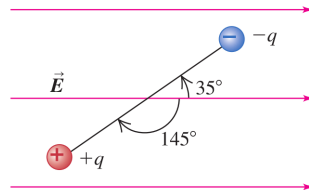
(b)



Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

(a)



(b)

